

STAT 452/652 Final Project Report

AUTHOR
Shivansh Ghai

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1. Introduction

The purpose of this project is to build an accurate predictive model for a continuous response Y using a training dataset of 20,000 observations and approximately 19 numeric predictors (X_1 – X_{19}). A separate test dataset containing only predictor values was provided. The goal was to generate predictions that minimize the **Mean Squared Prediction Error (MSPE)**, evaluated by the instructor.

Following the STAT 452/652 syllabus, I explored a wide range of linear, shrinkage, dimension-reduction, and ensemble methods. After comprehensive tuning and model comparison, a **Random Forest with $mtry = 11$ and $nodesize = 2$** achieved the strongest out-of-sample predictive performance and was selected as the final model.

2. What models did I try to use for this problem?

All models used fall strictly within the scope of STAT 452/652 (Lectures 1–15). They include:

Linear and Shrinkage Models

- Ordinary Least Squares (OLS)
- Ridge Regression
- LASSO Regression
- Elastic Net Regression

Dimension-Reduction Approaches

- Principal Components Regression (PCR)
- Partial Least Squares (PLS)

Tree-Based and Ensemble Methods

- Regression Trees (rpart)
- Random Forest (caret implementation)
- Random Forest (manual tuning using OOB error)
- Gradient Boosting Machines (manual gbm)
- Gradient Boosting Machines (caret implementation)

These models align with all material covered up to Lecture 15 and represent the full range of predictive tools recommended in the project instructions.

```
# Example: OLS model
ols_fit <- train(Y ~ ., data = train, method = "lm", trControl =
```

3. How did I evaluate and compare models?

All linear, shrinkage, and boosting methods were evaluated using **10-fold cross-validation**, as recommended in class. For Random Forest, **Out-of-Bag (OOB) MSE** was used, which serves as an internal cross-validation estimate.

Summary of Model Performance (Based on Code Output)

Model	CV RMSE / MSE	Notes
OLS	~5.19 RMSE	Baseline
Ridge	~5.22 RMSE	Slightly worse
LASSO	~5.19 RMSE	Comparable to OLS
Elastic Net	~26 MSE	Not competitive
PCR	~5.19 RMSE	No improvement
PLS	~5.19 RMSE	Similar

Regression Tree	~26 MSE	Underfits
Random Forest (caret)	4.01 RMSE	Strong performer
Tuned Random Forest (OOB)	15.84 MSE (~3.98 RMSE)	Best overall
GBM (caret)	4.44 RMSE	Good but weaker

The manually tuned **Random Forest** clearly outperformed all other models and therefore was chosen as the final predictive model.

```
ctrl <- trainControl(method = "cv", number = 10)
```

4. How did I choose those parameters?

Tuning procedures followed the course methods covered through Lecture 15.

Shrinkage Models

Ridge, LASSO, and Elastic Net were tuned over grids of **lambda** (and **alpha** for Elastic Net) using 10-fold CV. Best values were selected based on minimum RMSE.

PCR / PLS

Each method used cross-validation to determine the optimal number of components via RMSEP curves.

Regression Trees

The base rpart model was pruned using the **1-SE rule**, selecting a simpler subtree with the smallest cross-validated error plus one standard error.

Random Forest (caret)

A CV tuning grid evaluated $mtry \in \{3, 5, 7, 9, 11\}$. The best CV RMSE occurred at **mtry = 11**.

Random Forest (manual tuning)

A more exhaustive grid search evaluated combinations of **mtry** and **nodesize** using OOB MSE. The optimal values were: - **mtry = 11** - **nodesize = 2**

These were stored programmatically:

```
best_mtry <- 11
best_nodesize <- 2
```

These stored values were then inserted directly into the final model-fitting code block to ensure perfect reproducibility:

```
final_rf <- randomForest(
  x = X,
  y = y,
  mtry = best_mtry,
  ntree = 1500,
  nodesize = best_nodesize,
  importance = TRUE
)
```

Gradient Boosting (GBM)

Both manual and caret implementations tested combinations of: - Interaction depth - Shrinkage values - Number of trees - Minimum observations per node

Despite good performance (~4.44 RMSE), GBM was inferior to the tuned Random Forest.

5. How many true predictors do I think there are?

Two primary tools were used: - **LASSO coefficient paths** (variables shrunk to 0) - **Random Forest importance scores**

LASSO retained 17 predictors, though several had negligible magnitude. Random Forest importance plots revealed a smaller core set driving prediction, including: **X11, X13, X1, X8, X12, X5, X15, X9, X19**.

Combining both sources, a reasonable estimate is that **8–12 predictors** are truly influential in generating Y.

6. Final Model and Test Predictions

Below is the exact final model used to generate predictions in the required format:

```
set.seed(452)
library(randomForest)

train <- read.csv("training_data.csv")
test  <- read.csv("test_predictors.csv")

X <- train[, grep("^X", names(train))]
y <- train$Y

best_mtry <- 11
best_nodesize <- 2

final_rf <- randomForest(
  x = X,
  y = y,
  mtry = best_mtry,
  ntree = 1500,
  nodesize = best_nodesize,
  importance = TRUE
)

final_predictions <- predict(final_rf, newdata = test)

write.table(
  final_predictions,
  "my_predictions.csv",
  sep = ",",
  row.names = FALSE,
  col.names = FALSE
)
```

This code produces the exact predictions submitted.

7. Conclusion

Across all evaluated methods, the tuned **Random Forest** achieved the strongest predictive accuracy, outperforming linear, shrinkage, dimension-reduction, and boosting approaches. Its low OOB MSE and stable CV behavior made it the best choice for final predictions. The modeling process adhered strictly to STAT 452/652 methodology, and the final `.qmd` file guarantees full reproducibility of results.